A MORPHISM OF SIMPLICIAL SITES INDUCES A MORPHISM OF THE TOTAL TOPOI COMPATIBLE WITH THE COMPONENT INCLUSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two 'immediate' verifications come with their inputs in-span: (g'_n)^{-1} h_{total,*} = h_{n,*} g_n^{-1} is the section computation over objects of C'_n described at source line 1298, and the case-B adjoint equality (lines 1340-1348) follows from the explicit shriek formulas of lemma-restriction-to-components-site (card L46149 of this batch) together with the base-change identity h_n^{-1}(f'_varphi)^{-1} = f_varphi^{-1} h_m^{-1}; the author's spellings are preserved verbatim.

Editorial note. The label lemma-morphism-simplicial-sites names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The morphism-of-simplicial-sites datum. Same chapter, Remark \label{remark-morphism-simplicial-sites}. Lines 1276-1278: the componentwise morphisms h_n with their compatibility data.
- Sites, continuous functors and the associated adjoint pair. Sites chapter, Sections \ref{sites-section-continuous-functors} and \ref{sites-section-morphism-sites}. Lines 1292-1296: the functors v_{total,s} and v_{total}^p = v_{total}^s forming the adjoint pair.
- Sheaves on the total site as systems with transition maps. Same chapter, \label{lemma-describe-sheaves-simplicial-site-site}. Lines 1299-1304: a sheaf on C_total is a system (F_n, F(varphi)); used again for the exactness conclusion.
- Sites, cocontinuous functors and composition (case B). Sites chapter, \label{sites-lemma-composition-cocontinuous}. Lines 1334-1338: the morphism of topoi associated to the cocontinuous v_total and the diagram commutativity.
- Explicit description of the component shriek functors. Same chapter, \label{lemma-restriction-to-components-site} (card L46149 of this batch). Lines 1342-1348: the case-B adjoint transfer reads off the explicit coproduct formulas.

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Source reference. Original reference 09WF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at proof line(s) 1316; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 1345; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WD: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 844–844. Prerequisite pointer invoked at proof line(s) 1335; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WC: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 797–797. Prerequisite pointer invoked at proof line(s) 1300; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D95: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1247–1247. Prerequisite pointer invoked at proof line(s) 1278; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03L5: sources/stacks-chow/tags.txt, source lines 4701–4701. Prerequisite pointer invoked at proof line(s) 1338; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00WU: sources/stacks-chow/tags.txt, source lines 1196–1196. Prerequisite pointer invoked at proof line(s) 1307; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00X0: sources/stacks-chow/tags.txt, source lines 1202–1202. Prerequisite pointer invoked at proof line(s) 1308; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1276; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}_n, f_\varphi, u_\varphi$ and $\mathcal{C}'_n, f'_\varphi, u'_\varphi$ be as in Situation 09WE. Let h be a morphism between simplicial sites as in Remark 0D95. Then we obtain a morphism*

of topoi

$$h_{total} : Sh(\mathcal{C}_{total}) \to Sh(\mathcal{C}'_{total})$$

and commutative diagrams

$$\begin{array}{ccc} Sh(\mathcal{C}_n) & \xrightarrow[h_n]{} & Sh(\mathcal{C}'_n) \\ {\scriptstyle g_n}\downarrow & & \downarrow{\scriptstyle g'_n} \\ Sh(\mathcal{C}_{total}) & \xrightarrow{h_{total}} & Sh(\mathcal{C}'_{total}) \end{array}$$

Moreover, we have $(g'_n)^{-1} \circ h_{total,*} = h_{n,*} \circ g_n^{-1}$.

Proof. Case A. Say h_n corresponds to the continuous functor $v_n : \mathcal{C}'_n \to \mathcal{C}_n$. Then we can define a functor $v_{total} : \mathcal{C}'_{total} \to \mathcal{C}_{total}$ by using v_n in degree n. This is clearly a continuous functor (see definition of coverings in Lemma 09WC). Let $h_{total}^{-1} = v_{total,s} : Sh(\mathcal{C}'_{total}) \to Sh(\mathcal{C}_{total})$ and $h_{total,*} = v_{total}^s = v_{total}^p : Sh(\mathcal{C}_{total}) \to Sh(\mathcal{C}'_{total})$ be the adjoint pair of functors constructed and studied in Sites, Sections 00WU and 00X0. To see that h_{total} is a morphism of topoi we still have to verify that h_{total}^{-1} is exact. We first observe that $(g'_n)^{-1} \circ h_{total,*} = h_{n,*} \circ g_n^{-1}$; this is immediate by computing sections over an object U of $\mathcal{C}'_n$. Thus, if we think of a sheaf $\mathcal{F}$ on $\mathcal{C}_{total}$ as a system $(\mathcal{F}_n, \mathcal{F}(\varphi))$ as in Lemma 09WF, then $h_{total,*}\mathcal{F}$ corresponds to the system $(h_{n,*}\mathcal{F}_n, h_{n,*}\mathcal{F}(\varphi))$. Clearly, the functor $(\mathcal{F}'_n, \mathcal{F}'(\varphi)) \to (h_n^{-1}\mathcal{F}'_n, h_n^{-1}\mathcal{F}'(\varphi))$ is its left adjoint. By uniqueness of adjoints, we conclude that h_{total}^{-1} is given by this rule on systems. In particular, h_{total}^{-1} is exact (by the description of sheaves on $\mathcal{C}_{total}$ given in the lemma and the exactness of the functors h_n^{-1}) and we have our morphism of topoi. Finally, we obtain $g_n^{-1} \circ h_{total}^{-1} = h_n^{-1} \circ (g'_n)^{-1}$ as well, which proves that the displayed diagram of the lemma commutes.

Case B. Here we have a functor $v_{total} : \mathcal{C}_{total} \to \mathcal{C}'_{total}$ by using v_n in degree n. This is clearly a cocontinuous functor (see definition of coverings in Lemma 09WD). Let h_{total} be the morphism of topoi associated to v_{total}. The commutativity of the displayed diagram of the lemma follows immediately from Sites, Lemma 03L5. Taking left adjoints the final equality of the lemma becomes

$$h_{total}^{-1} \circ (g'_n)_!^{Sh} = g_{n!}^{Sh} \circ h_n^{-1}$$

This follows immediately from the explicit description of the functors $(g'_n)_!^{Sh}$ and $g_{n!}^{Sh}$ in Lemma 09WG, the fact that $h_n^{-1} \circ (f'_\varphi)^{-1} = f_\varphi^{-1} \circ h_m^{-1}$ for $\varphi : [m] \to [n]$, and the fact that we already know h_{total}^{-1} commutes with restrictions to the degree n parts of the simplicial sites. □